

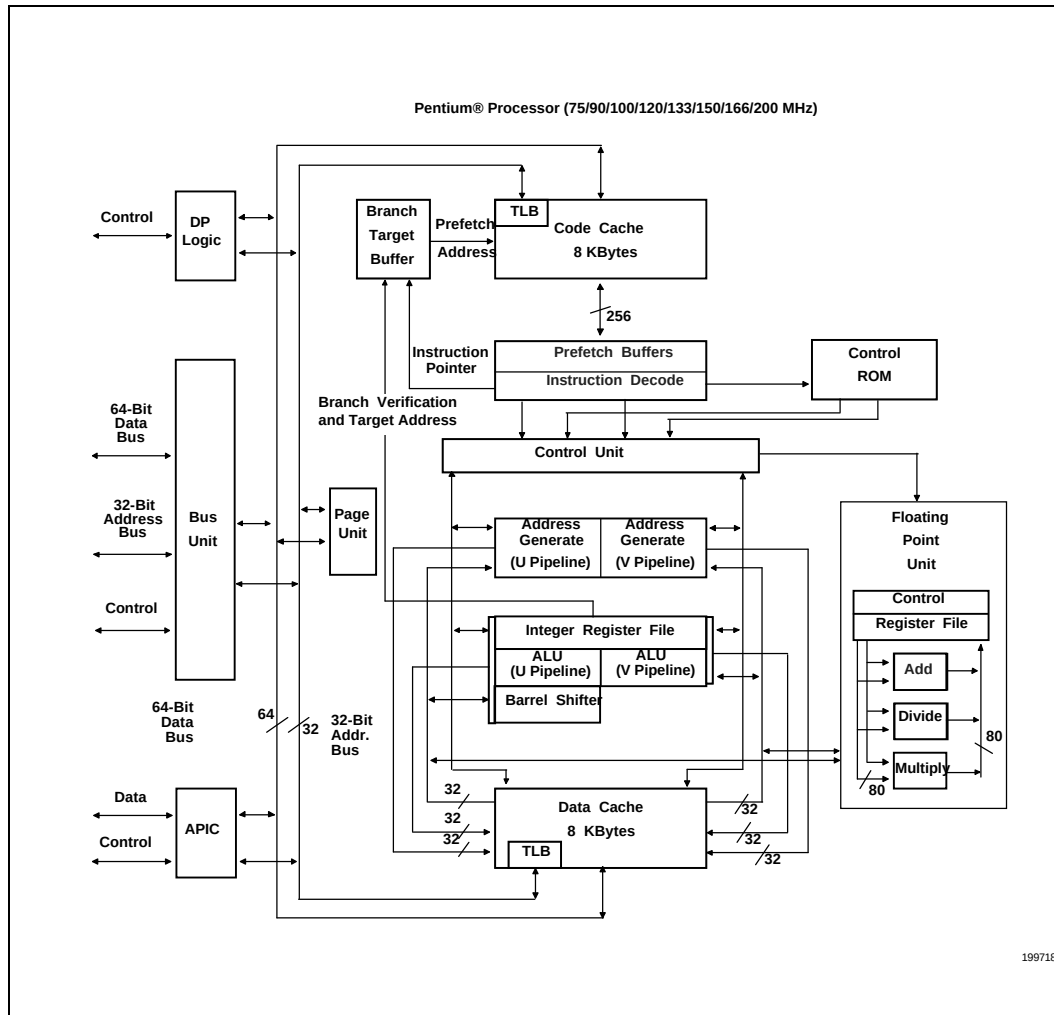


Figure 1. Pentium® Processor Block Diagram

The block diagram shows the two instruction pipelines, the "u" pipe and the "v" pipe. The u-pipe can execute all integer and floating point instructions. The v-pipe can execute simple integer instructions and the FXCH floating-point instructions.

The separate code and data caches are shown. The data cache has two ports, one for each of the two pipes (the tags are triple ported to allow simultaneous inquire cycles). The data cache has a dedicated Translation Lookaside Buffer (TLB) to

translate linear addresses to the physical addresses used by the data cache.

The code cache, branch target buffer and prefetch buffers are responsible for getting raw instructions into the execution units of the Pentium processor. Instructions are fetched from the code cache or from the external bus. Branch addresses are remembered by the branch target buffer. The code cache TLB translates linear addresses to physical addresses used by the code cache.